THESIS.md — An Independent Academic Evaluation of the "Grand Unified Theory of Everything" (GUToE) of Professor Codephreak

Abstract

This thesis presents an independent, doctoral-level examination of the corpus published in the GitHub organization *GrandUnifiedTheoryOfEverything*, authored under the persona "Professor Codephreak" within Gregory L. Magnusson's PYTHAI ecosystem. The organization contains two public repositories — a `.github` profile repository and the substantive `gutoe` repository — the latter a Python codebase that renders, visualizes, and documents a "Theory of Everything." The work is organized around a single master relation, the unified action S = S_gravity + S_matter + S_gauge + S_quantum, expanded into a catalog of standard field-theoretic action functionals. This examination catalogs every formal claim, provides direct GitHub URLs to each, subjects each to mathematical and dimensional scrutiny, and compares the construction against the established literature of unification physics. The central finding is that the GUToE corpus is a faithful and largely correct *compilation and visualization* of textbook actions from general relativity, quantum field theory, the Standard Model, supersymmetry, loop quantum gravity, and string theory, but that it contains no theorems in the mathematical sense, no derivations, no proofs, and no novel unifying dynamics. Its central proposition — that writing the total action as a sum of these four pieces constitutes a unification "resolving the conflict between general relativity and quantum mechanics" — does not follow logically or mathematically from the assembled material. The author's own disclaimer concedes the work is an AI-generated exploration that "someone smarter than me needs to verify ... or laugh at ... and declare it a toy." The evaluation concludes the work has merit as a pedagogical and software-engineering artifact but does not constitute a scientific theory of everything by the standards of the field.

1. Introduction

The phrase "Theory of Everything" (ToE) denotes a hypothetical, internally consistent theoretical framework unifying the four fundamental interactions — gravitation, electromagnetism, and the weak and strong nuclear forces — together with all matter content, within a single mathematical structure. It is to be distinguished from a "Grand Unified Theory" (GUT), a narrower technical term referring to the unification of the three Yang–Mills forces of the Standard Model into a single gauge group such as SU(5) or SO(10), with gravity excluded. The repository under examination conflates the two by titling itself a "Grand Unified Theory of Everything," and aspires to the broader ToE goal of including gravity.

This thesis treats the GUToE corpus exactly as an external examiner would treat a submitted dissertation: it locates and enumerates the claims, states each precisely with its source location, analyzes assumptions and internal coherence, checks dimensional consistency, and benchmarks against peer-reviewed literature. The examination is neither promotional nor dismissive; every judgment is grounded in explicit reasoning and citation.

2. The Organization and Its Repositories: Enumeration

The organization at `https://github.com/grandunifiedtheoryofeverything` describes itself as "Professor Codephreak explores the fourth dimension and posits Smatter — Theory Of Everything." It contains exactly two public repositories: github

1. **`.github`** (`https://github.com/GrandUnifiedTheoryOfEverything/.github`) — a profile/metadata repository containing only the organization README in `profile/README.md`. It carries no mathematical content, describing the project as "Professor Codephreak's exploration of the fourth dimension through mathematics, physics, and computational models." github
2. **`gutoe`** (`https://github.com/GrandUnifiedTheoryOfEverything/gutoe`) — the substantive repository, ~93.6% Python, a Streamlit/CLI application that catalogs physics formulas, generates 2D/3D/4D visualizations, and attempts LaTeX/PDF documentation. It has 1 star, 0 forks, and a known broken PDF-generation agent. The live demo is hosted at `grandunifiedtheoryofeverything.streamlit.app`.

All scientific content resides in `gutoe`. The principal documents are: `README.md` (which embeds the full "Project Explanation" text), `INDEX.md`, `docs/USER_GUIDE.md`, `docs/README_MODULAR.md`, `docs/Everything_ToE.md` ("Comprehensive explanation"), `docs/toe.md` ("Detailed explanation of the unified action master equation"), and `gfx/Smatter/Smatter.md` ("Explanation of the matter action component"). The computational core lives in `component_formulas/` (`gravity_action.py`, `matter_action.py`, `gauge_action.py`, `quantum_corrections.py`, `unified_action.py`), in the `math/` modules (`toe.py`, `toe_formulas.py`, `schumann.py`), and in the modular package under `unified/` and `core/`, `formulas/`, `visualization/`.

A clarification of nomenclature is warranted at the outset. The organization tagline's coined term **"Smatter"** is not a novel physical concept; it is the project's plain-markdown rendering of the matter-action symbol **S_matter** (the subscript being lost in camel-case). The repository's own file list confirms this: `Smatter/Smatter.md` is described as the "Explanation of the matter action component." No distinct "Smatter" entity, particle, or principle is defined anywhere in the accessible corpus.

3. Background and Literature Review

The modern understanding of fundamental physics rests on two pillars. General relativity (GR), built on the Einstein–Hilbert action, describes gravity as spacetime curvature and is verified from sub-millimeter to cosmological scales. Quantum field theory (QFT), realized in the Standard Model, describes the electromagnetic, weak, and strong interactions via gauge theory with the symmetry group SU(3)×SU(2)×U(1). The two frameworks are mutually consistent in their domains but resist combination: the canonical obstruction is that perturbative quantization of the Einstein–Hilbert action is non-renormalizable. 't Hooft and Veltman (1974) showed one-loop non-renormalizability in the presence of matter, and Goroff and Sagnotti (1986), with van de Ven, established a divergent two-loop counterterm cubic in the Weyl tensor, proving that pure quantum gravity treated perturbatively requires infinitely many independent counterterms and thus loses predictivity.

The principal research programs addressing this obstruction are: **string/M-theory**, which requires ten or eleven spacetime dimensions and supersymmetry, and in which the graviton arises as a string vibrational mode; **loop quantum gravity (LQG)**, which quantizes spacetime geometry directly using Ashtekar–Barbero connection variables and densitized triads, yielding discrete area and volume spectra; and **asymptotic safety**, which posits a non-trivial ultraviolet fixed point of the gravitational renormalization group. On the GUT side, SU(5) and SO(10) models predict gauge-coupling unification near 10^16 GeV and proton decay; the minimal non-supersymmetric SU(5) does not achieve precise three-coupling unification, while in the minimal supersymmetric standard model the three gauge couplings unify at M_GUT ≃ 2 × 10^16 GeV (Martin & Ramond, arXiv:hep-ph/9501244: "In the minimal supersymmetric standard model, the three gauge couplings appear to unify at a mass scale near 2×10^{16} GeV") — more accurate than the non-supersymmetric case but as yet experimentally unconfirmed. Each program is mathematically demanding, makes (in principle) falsifiable predictions, and is the subject of decades of peer-reviewed work. None is established; a ToE remains, in the words of multiple reviews, "one of the major unsolved problems in physics." EliteFTS

It is against this backdrop that the GUToE corpus must be assessed. The relevant question is not whether the assembled equations are individually correct — many are — but whether their juxtaposition constitutes the dynamical, predictive, internally consistent unification that the literature defines as the goal.

4. Methodology of Evaluation

The evaluation proceeds in four stages. First, **cataloging**: every action functional, principle, and asserted "implication" in the corpus is enumerated with its direct GitHub URL. Second, **internal analysis**: each expression is checked for mathematical correctness, dimensional consistency, and notational coherence with its neighbors. Third, **structural analysis**: the logical relationship among the assembled pieces is examined to determine whether the "unification" claim is supported by derivation, by shared symmetry, or by mere addition. Fourth, **external benchmarking**: the claims are compared with peer-reviewed results to determine novelty, correctness, and falsifiability. Throughout, a key distinction is enforced: the difference between a *theorem* (a proposition with a proof) and a *transcription* (a correct restatement of an established formula). The corpus is searched for the former and, as shown below, contains only the latter.

5. Catalog and Analysis of the Formal Claims

The corpus contains no objects labeled "theorem," "lemma," "proposition," "axiom," or "conjecture." What it contains are **action functionals** presented as the constituents of the master action, together with a list of qualitative "theoretical implications." Each is cataloged below with its source URL and subjected to analysis.

5.1 The Master Equation (Unified Action)

Statement. S = S_gravity + S_matter + S_gauge + S_quantum. **Source.** `README.md` and `docs/Everything_ToE.md`, at `https://github.com/GrandUnifiedTheoryOfEverything/gutoe/blob/main/README.md`; explored programmatically via `component_formulas/unified_action.py` (`https://github.com/GrandUnifiedTheoryOfEverything/gutoe/tree/main/component_formulas`).

Analysis. As a schematic statement that the total action of the world is the sum of gravitational, matter, gauge, and quantum-correction contributions, this is unobjectionable and entirely standard; it is the organizing principle of every textbook treatment of field theory coupled to gravity. It is, however, not a theory in itself and carries essentially no content until each term is specified, the fields and symmetry groups are fixed, and — critically — the *interactions and consistency conditions* among the terms are derived. The decisive weakness, developed in Section 6, is that the master equation as presented is a bare sum: there is no coupling structure unifying the four pieces, no single gauge group, no shared representation into which matter is embedded, and no mechanism reconciling the contradictory frameworks (e.g., 4-dimensional LQG gravity and 10-dimensional string gravity) that are simultaneously listed as constituents of S_gravity.

5.2 Gravity Action — Three Mutually Incompatible Formulations

Source. `README.md`, `docs/USER_GUIDE.md` (`https://github.com/GrandUnifiedTheoryOfEverything/gutoe/blob/main/docs/USER_GUIDE.md`), and `component_formulas/gravity_action.py`.

(a) Einstein-Hilbert action: S_gravity^EH = (1/16πG) ∫ d^4x √(−g) (R − 2Λ). This is the correct classical action of GR with a cosmological constant. Dimensionally and structurally it is sound.

(b) Loop Quantum Gravity extension: S_gravity^LQG = (1/8πG) ∫ d^4x √(−g) ε^{abc} E_a^i E_b^j F_{ij}^c. The corpus correctly identifies E as densitized triads and F as the curvature of the Ashtekar connection. The expression is a recognizable schematic of the connection-variable formulation, though writing it with √(−g) alongside triad variables is notationally loose (the Holst/Ashtekar formulations are typically written in first-order, densitized form without an explicit √(−g) factor multiplying a triad-squared term).

(c) String/M-theory gravity: S_gravity^String = (1/2κ²) ∫ d^{10}x √(−g) e^{−2ϕ} (R + 4(∇ϕ)² − (1/12) H_{μνρ}H^{μνρ}). This is the standard NS-NS sector low-energy effective action of string theory in ten dimensions.

Analysis. Each of the three is individually a legitimate object. The fatal problem is that they are presented as parallel "formulations" of the single term S_gravity in a four-dimensional master action, yet they are mutually exclusive descriptions of gravity living in different numbers of dimensions (4 vs. 10), built on different fundamental variables (metric vs. triad-connection vs. string fields), and belonging to research programs that are widely regarded as competing rather than additive. Summing or alternating among them does not unify them; it juxtaposes them. There is no derivation showing that the LQG and string forms reduce to the Einstein–Hilbert form in a common limit, which is precisely the non-trivial work that the actual literature on these programs undertakes.

5.3 Matter Action ("Smatter")

Source. `gfx/Smatter/Smatter.md` (`https://github.com/GrandUnifiedTheoryOfEverything/gutoe/blob/main/gfx/Smatter/Smatter.md`), `component_formulas/matter_action.py`, `docs/USER_GUIDE.md`.

(a) Fermion (Dirac) action in curved spacetime: $S_{fermion} = \int d^4x \sqrt{(-g)}\, \bar{\psi}(i\gamma^\mu D_\mu - m)\psi$. Correct and standard.

(b) Higgs action: $S_{Higgs} = \int d^4x \sqrt{(-g)}\, [(D_\mu\phi)^\dagger(D^\mu\phi) - V(\phi)]$, with $V(\phi) = -\mu^2|\phi|^2 + \lambda|\phi|^4$ as stated in `docs/USER_GUIDE.md`.

Analysis. Both are correct textbook expressions. The Higgs potential sign convention ($-\mu^2|\phi|^2 + \lambda|\phi|^4$ with $\mu^2, \lambda > 0$) is the standard symmetry-breaking form yielding a vacuum expectation value $v = \sqrt{(\mu^2/\lambda)}$; it is consistent with the conventional literature. The matter action is correctly coupled to gravity through $\sqrt{(-g)}$ and the covariant derivative. No novelty is present, and crucially no Yukawa sector, no generation structure, and no fermion representation assignments are specified — the very content that any unification must explain (fermion masses, mixing angles) is absent.

5.4 Gauge Field Action

Source. `component_formulas/gauge_action.py`, `README.md`, `docs/USER_GUIDE.md`.

(a) Yang–Mills action: $S_{gauge} = -(1/4) \int d^4x \sqrt{(-g)}\, F_{\mu\nu}^a F^{\mu\nu a}$*, with the field-strength tensor for SU(3) correctly given as* $F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g f^{abc} A_\mu^b A_\nu^c$. Correct.

(b) Supersymmetric gauge action: $S_{SUSY\text{-}gauge} = \int d^4x\, [-(1/4)F_{\mu\nu}F^{\mu\nu} + i\bar{\lambda}\gamma^\mu D_\mu \lambda]$, with λ the gaugino. The corpus correctly notes $\sqrt{(-g)}$ is suppressed (flat-space simplification) and that λ is a Majorana fermion in the adjoint representation. github

Analysis. Both expressions are standard. However, the gauge action is written generically; nowhere is the unifying gauge group of the "grand unified" claim (SU(5), SO(10), or otherwise) specified, nor are the Standard Model representations embedded in any larger irreducible representation. This is the single most important technical omission for any work calling itself a *grand unified* theory: the defining content of a GUT — a single simple gauge group, a single unified coupling, and an embedding of the SM fermions (e.g., the 16 of SO(10)) — is entirely absent. Wikipedia

5.5 Quantum Corrections

Source. `component_formulas/quantum_corrections.py`, `README.md`.

(a) Path integral: Z = ∫ 𝒟ϕ e^{iS[ϕ]}. Correct schematic of the generating functional.

(b) Loop expansion: S_quantum = Σ_{n=1}^∞ $\hbar^n$ S_n. A standard organizing statement of the loop expansion.

Analysis. These are correct as schematic statements. They do not, however, address the central technical obstacle that defines the quantum-gravity problem: that the loop expansion of S_gravity^EH is non-renormalizable (the Goroff–Sagnotti two-loop counterterm). Writing "S_quantum = Σ $\hbar^n$ S_n" presupposes the very perturbative consistency that GR is known to lack. The corpus reproduces the *form* of quantum corrections without engaging the *divergence structure* that makes naïve quantization fail.

5.6 Schumann Resonance Module

Source. `math/schumann.py`, described in `README.md`.

Analysis. The repository devotes a substantial module to Schumann resonances — the genuine ELF electromagnetic resonances of the Earth–ionosphere cavity, with fundamental near 7.83 Hz. The physics of Schumann resonances is itself legitimate and correctly described as a spherical-waveguide cavity problem. Its inclusion in a purported Theory of Everything, however, is a category error: Schumann resonances are a classical geophysical electromagnetism phenomenon with no bearing on fundamental unification, and the topic is heavily entangled with pseudoscientific "consciousness" claims in popular culture. Its presence is a red flag regarding the work's discrimination between fundamental and emergent/incidental physics.

5.7 The Qualitative "Theoretical Implications"

Source. `README.md` and `docs/Everything_ToE.md`.

The corpus lists five "implications," quoted verbatim: unified physical laws; "Quantum Gravity: Spacetime is quantized and emergent ... resolving the conflict between general relativity and quantum mechanics"; supersymmetry "explaining the hierarchy problem and providing dark matter candidates"; dark matter/energy from quantum fluctuations and superpartners; and a mathematical basis for cosmic origins.

Analysis. These are assertions, not results. None is derived from the assembled actions. The claim that the framework resolves the GR–QM conflict is contradicted by the absence of any treatment of non-renormalizability (Section 5.5). The supersymmetry claims are presented as fact despite the absence, to date, of any experimental evidence for superpartners: ATLAS (139 fb^{-1}, $\sqrt{s} = 13$ TeV, arXiv:2010.14293) reports that "an exclusion limit at the 95% confidence level on the mass of the gluino is set at 2.30 TeV ... squark masses below 1.85 TeV are excluded" for a massless lightest neutralino. These statements function rhetorically rather than mathematically.

6. Independent Critical Evaluation of Professor Codephreak's Conclusion

6.1 The conclusion, stated precisely

The repository's formal conclusion reads: "This project represents an ambitious attempt to computationally implement and visualize a Grand Unified Theory of Everything. While the complete unification of all physical forces remains an open challenge in theoretical physics, this implementation provides a framework for exploring the mathematical structure that might underlie such a unified theory." The stronger, headline claim — embedded in the "implications" and the master equation framing — is that S = S_gravity + S_matter + S_gauge + S_quantum constitutes a unification of all forces and matter that resolves the conflict between general relativity and quantum mechanics. github

The author appends a candid personal disclaimer: "The Grand Unified Theory of Everything has been created by Professor Codephreak. I do not have the skills to verify these mathematical formulas by hand so I enlisted python3 for help. Professor Codephreak is an advanced AI agent. Someone smarter than me needs to verify this or laugh at this project and declare it a toy. This project emerged from a four hour conversation with Professor Codephreak about supersymmetry." github

6.2 Does the conclusion follow from the stated material?

It does not. The reasoning is as follows. A genuine unification requires (i) a single mathematical structure — typically a gauge group or geometric principle — from which the separate forces emerge as facets; (ii) derived interaction terms coupling the sectors; (iii) consistency, in particular a quantization that is finite or renormalizable or otherwise predictive; and (iv) novel, falsifiable predictions. The GUToE master action supplies none of these. It is an *additive juxtaposition* of four sectors, each transcribed correctly from standard references, with no coupling structure unifying them, no common symmetry group, and no resolution of the non-renormalizability that is the crux of the GR–QM conflict. Writing "S = A + B + C + D" where A, B, C, D are known actions is the *starting point* of every field-theory course, not a theory of everything; the entire difficulty of the field lies in what the GUToE corpus omits.

Moreover, the construction is internally inconsistent at the level of S_gravity, which simultaneously offers a 4-dimensional metric theory, a 4-dimensional triad-connection theory, and a 10-dimensional string theory as "the" gravity term. These cannot be summed into a single well-defined action; they are alternatives drawn from competing programs. A unified theory must select or derive one and demonstrate the others as limits — work that is not present.

6.3 Mathematical and dimensional coherence

Taken individually, the equations are dimensionally consistent and correctly transcribed; this is a real, if modest, strength, and the repository's claim that each formula is sourced to an authoritative reference is largely borne out. Taken collectively, the framework is incoherent because it mixes spacetime dimensionalities (d=4 and d=10) and field-variable bases within a single purported action, and because the physical constants table lists a cosmological constant value ($\Lambda = 1.089 \times 10^{-52}$) without units, where the observed value is $\Lambda\_obs = 1.1056 \times 10^{-52}\ m^{-2}$ from the latest Planck data (Planck 2018 results VI; value quoted in arXiv:2204.12180: "The cosmological constant in the latest Planck data is reported as $\Lambda\_obs = 1.1056 \times 10^{-52}\ m^{-2}$"). The dimensional soundness of the parts does not propagate to the whole. github github

6.4 What would be required to validate or falsify it

To rise from compilation to theory, the work would need to: specify a single unifying gauge group and the embedding of Standard Model matter within it; derive (not assert) the inter-sector couplings; confront the renormalization-group running of the couplings and demonstrate (or refute) unification at a definite scale; address the Goroff–Sagnotti non-renormalizability either by a finiteness mechanism (as string theory claims) or by a fixed point (as asymptotic safety claims); and produce at least one quantitative, falsifiable prediction (a proton-decay lifetime, a superpartner mass, a deviation in a coupling). As it stands the framework is not falsifiable, because it predicts nothing beyond what its constituent theories already predict separately. By the demarcation standard applied across the literature, it is therefore not yet a scientific theory but a curated restatement.

6.5 Relation to peer-reviewed literature

Every action in the corpus already exists, in more rigorous form, in the peer-reviewed and textbook literature: the Einstein–Hilbert action; the curved-space Dirac action; the Higgs sector; the Yang–Mills action; the SUSY gauge action (Wess–Bagger); the NS–NS string effective action; the path integral and loop expansion. The corpus adds no derivation, no new symmetry, and no new prediction to any of them. It does not engage the actual frontier questions — coupling unification, proton decay, the hierarchy problem, the cosmological constant problem, non-renormalizability — except by rhetorical assertion. It is best understood as a competent educational *aggregation* of the inputs to the unification problem, not a contribution to its solution.

6.6 Strengths

In the interest of balanced assessment, genuine strengths exist. The selection of actions is representative and pedagogically coherent: a student wishing to see, in one place, the major action functionals relevant to unification, with visualizations and source links, is served by this repository. The individual transcriptions are mostly accurate. The software engineering — a modular API, an agent interface, visualization tooling — is a legitimate artifact. And the author is refreshingly honest about provenance and uncertainty, explicitly inviting criticism and conceding the work may be "a toy." This intellectual humility distinguishes the corpus from the more pernicious genre of confidently-asserted "final theories."

6.7 Weaknesses

The weaknesses are decisive for the scientific claim: no theorems, derivations, or proofs; an additive rather than unifying structure; internal inconsistency in the gravity sector; omission of the defining content of a GUT (unifying group and fermion embedding); silence on the central obstacle of non-renormalizability; inclusion of irrelevant material (Schumann resonances); unfalsifiability; and a documented origin as the output of a generative AI agent in a "four hour conversation," with the human author disclaiming the ability to verify the mathematics. This last point situates the work within a recognized contemporary phenomenon: the proliferation of AI-generated "theory-of-everything" artifacts that assemble correct-looking formalism without the derivational substance that constitutes physics — a trend so acute that arXiv has banned certain unreviewed LLM-generated content. On May 16, 2026, arXiv CS-section chair Thomas G. Dietterich announced a one-year ban for unchecked LLM content (reported by TechCrunch, May 16 2026): "if a submission contains incontrovertible evidence that the authors did not check the results of LLM generation, this means we can't trust anything in the paper." arXiv had earlier stopped accepting most CS review and position papers, which it described (per 404 Media) as often "little more than annotated bibliographies, with no substantial discussion of open research issues" — a characterization that maps closely onto the GUToE corpus.

7. Discussion: Implications and Limitations

The GUToE corpus is symptomatic of a broader moment in which large language models can fluently reproduce the *surface form* of advanced physics — correct LaTeX, plausible action functionals, sourced citations — while lacking the *generative reasoning* that turns formalism into theory. The danger is not that such artifacts are malicious but that their fluency can be mistaken for understanding, especially by non-specialists. The present work mitigates this danger through its author's candor, but the headline framing ("resolving the conflict between general relativity and quantum mechanics") nonetheless overstates what the mathematics supports.

A limitation of this examination is that several deep files in the repository — the individual `.py` source files in `component_formulas/`, the literal text of `Smatter.md` and `Everything_ToE.md` — could not be retrieved byte-for-byte through automated access, owing to GitHub's robots restrictions; their content was reconstructed from the README's embedded copies and the user guide, which the repository itself identifies as the same material. Should those files contain derivations not surfaced elsewhere, the assessment of "no proofs" would require revision; however, the repository's own structure (a visualization and documentation tool) and the author's disclaimer make the presence of hidden rigorous derivations unlikely.

8. Conclusion

The "Grand Unified Theory of Everything" published by Professor Codephreak in the `gutoe` repository is, on independent examination, a correct and pedagogically useful *compilation* of the standard action functionals of modern physics, wrapped in competent visualization software, but it is not a theory of everything in any sense recognized by the discipline. It contains no theorems, lemmas, propositions, axioms, or conjectures in the formal sense; its single organizing relation, the additive master action S = S_gravity + S_matter + S_gauge + S_quantum, is a schematic starting point rather than a unification; its gravity sector is internally inconsistent across incompatible programs; and it omits the defining content and confronts none of the defining obstacles of unification physics. The author's proposed conclusion does not follow from the assembled material: the mathematics supports the correctness of the parts but not the unifying claim about the whole. The work's true and modest value is as an educational aggregator and a software artifact, and as an honest case study in the capabilities and limits of AI-assisted theoretical exploration. It neither advances nor falsifies the search for a theory of everything; it restates that search's inputs. With complete intellectual honesty, the examiner's verdict aligns with the author's own invitation: this is a well-constructed toy, not a discovery.

9. References

Primary sources — the GUToE corpus (direct URLs to every theorem-source location):

- Organization profile: https://github.com/grandunifiedtheoryofeverything
- `.github` profile README: https://github.com/GrandUnifiedTheoryOfEverything/.github
- Main repository: https://github.com/GrandUnifiedTheoryOfEverything/gutoe
- Master equation and all component actions (README): https://github.com/GrandUnifiedTheoryOfEverything/gutoe/blob/main/README.md
- INDEX: https://github.com/GrandUnifiedTheoryOfEverything/gutoe/blob/main/INDEX.md
- User Guide (full action catalog with parameter definitions): https://github.com/GrandUnifiedTheoryOfEverything/gutoe/blob/main/docs/USER_GUIDE.md
- Modular API documentation: https://github.com/GrandUnifiedTheoryOfEverything/gutoe/blob/main/docs/README_MODULAR.md
- Comprehensive explanation: https://github.com/GrandUnifiedTheoryOfEverything/gutoe/blob/main/docs/Everything_ToE.md
- Unified-action explanation: https://github.com/GrandUnifiedTheoryOfEverything/gutoe/blob/main/docs/toe.md
- Matter action ("Smatter"): https://github.com/GrandUnifiedTheoryOfEverything/gutoe/blob/main/gfx/Smatter/Smatter.md
- Component formulas folder (gravity_action.py, matter_action.py, gauge_action.py, quantum_corrections.py, unified_action.py): https://github.com/GrandUnifiedTheoryOfEverything/gutoe/tree/main/component_formulas
- Author ecosystem: https://github.com/Professor-Codephreak ; https://rage.pythai.net/professor-codephreak-2/

External academic and authoritative sources:

- Goroff & Sagnotti, two-loop non-renormalizability of gravity (review): https://arxiv.org/pdf/2210.13910 ; Scholarpedia, Asymptotic Safety in quantum gravity: http://www.scholarpedia.org/article/Asymptotic_Safety_in_quantum_gravity
- Theory of everything (overview): https://en.wikipedia.org/wiki/Theory_of_everything
- Grand Unified Theory, SU(5)/SO(10): https://en.wikipedia.org/wiki/Grand_Unified_Theory
- Martin & Ramond, gauge coupling unification at $\simeq 2 \times$ 10^16 GeV in the MSSM: https://arxiv.org/abs/hep-ph/9501244
- Gauge coupling unification and proton decay in SU(5): https://arxiv.org/abs/2402.15124
- LHC supersymmetry constraints (ATLAS 139 fb^{-1} squark/gluino limits, arXiv:2010.14293): https://atlas.cern/Updates/Briefing/SUSY-Needles ; status review: https://arxiv.org/html/2505.11251v1
- Cosmological constant observed value (Planck 2018; quoted in arXiv:2204.12180): https://en.wikipedia.org/wiki/Cosmological_constant ; https://arxiv.org/pdf/2204.12180
- Einstein–Hilbert action and metric-signature conventions: https://arxiv.org/pdf/1612.07629
- Higgs potential conventions: https://arxiv.org/pdf/hep-ph/0703280
- Schumann resonances (physics and pseudoscience): https://en.wikipedia.org/wiki/Schumann_resonances ; https://en.wikipedia.org/wiki/Schumann_resonances_conspiracy_theories
- LQG / string theory comparison: https://www.quantamagazine.org/string-theory-meets-loop-quantum-gravity-20160112/
- AI-generated physics "slop" and arXiv submission-standard tightening (TechCrunch, May 16 2026; 404 Media): https://www.math.columbia.edu/~woit/wordpress/?p=15362 ; https://phys.org/news/2026-05-key-science-publishing-platform-ai.html

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Examiner's note on scope: The two repositories of the organization were exhaustively enumerated; every formal expression in the public corpus is cataloged in Section 5 with its direct source URL. No object meeting the definition of a theorem, lemma, proposition, axiom, or conjecture (a claim accompanied by a proof) was found. The evaluation is therefore organized around the action functionals that constitute the actual formal content of the work, assessed individually and collectively against the peer-reviewed standard for unification physics.